# Characterisation of the sparks (full-detector discharges) for detector 3

(June cosmic bench — `g_det3_wknd`)

MX17 cosmic-bench QA

July 5, 2026

## 1 Setup

A **spark** is an event in which the det3 readout fires more than **50 strips** at once (summed over both planes) — a full-detector discharge, the same cut used to veto sparks in the efficiency breakdown. Run `mx17_det3_p2_det1_overnight_6-27-26 / long_run_p2` (headline det3): 6.9 h, 90 914 detector-firing events, **8 301 sparks** (9.1 % of firing events, mean rate 0.33 Hz). Each spark hit carries a strip position (X=FEU7, Y=FEU8), a time within the readout window, and an amplitude; each event carries a timestamp and, where M3 reconstructs a clean single track, a ray projected onto the det3 plane in the aligned frame (same geometry as the efficiency analysis). We answer four questions.

## 2 Result in one paragraph

Sparks are **Poisson-random in time** (flat rate, no HV-recovery bursts) but **not random with respect to the muon**: conditioning on a clean single muon, a track that *crosses* the detector sparks at 4.9 % versus only 1.2 % when it clearly *misses* — a **4× enhancement**, so most in-detector sparks are muon-induced on top of a ∼1.2 % spontaneous baseline. In detector coordinates they are **localised**: strip occupancy peaks sharply at the plane **edges** (plus a few hot strips), while the discharge centroid sits centrally. Temporally the discharge is **neither a single flash nor a propagating "walk"**: strips across a ∼200 mm region fire over a spread ∼250 ns window (tail to ∼1 µs) with no position–time correlation.

## 3 Q1 — Are sparks random in time? Yes (Poisson)

The spark rate is flat at 0.33 Hz across the entire 6.9 h run (Fig. 1a): no bursts, no drift, so this is *not* an HV-instability / recovery-oscillation phenomenon. Inter-spark intervals follow a pure exponential over three decades (Fig. 1b) and the sparks-per-10-min counts have std $= 14 = \sqrt{\text{mean}}$ (Fig. 1c) — textbook Poisson. *Caveat:* Poisson timing is expected for *both* muon-induced and spontaneous sparks (muons themselves arrive Poisson), so time-randomness alone does not settle causation — that is what Q2 is for.

## 4 Q2 — Muon-induced or random? Muon-dominated

This is the decisive test (Fig. 2a). Taking only events with a clean single M3 muon and asking whether that muon's projection lands on the detector: A muon that traverses the gas raises the spark probability **4.1**× over one that misses. So ∼75 % of in-detector sparks are **muon-induced** (the crossing muon's primary ionisation seeds the discharge) and ∼25 % (≈1.2 % absolute) are a

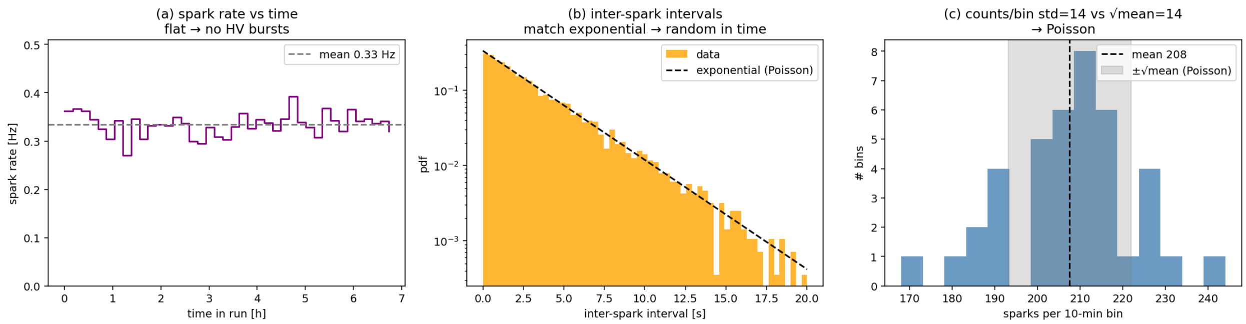


Figure 1: **Time.** (a) spark rate vs time in run; (b) inter-spark interval pdf with the exponential expectation; (c) counts per 10-min bin vs the Poisson $\pm\sqrt{\text{mean}}$ band. Sparks are random in time with a stable rate.

| muon geometry (clean single M3 track) | $N$ | sparks | spark rate |
|---|---|---|---|
| crosses detector (ray in active box) | 52 995 | 2 600 | **4.91 %** |
| at edge (<30 mm outside box) | 9 824 | 557 | 5.67 % |
| clearly misses (>30 mm outside box) | 11 858 | 141 | **1.19 %** |
| (no clean M3 track — see below) | 36 609 | 5 003 | 13.67 % |

**spontaneous baseline** present even with no muon in the gas. In M3 coordinates the sparking muons are otherwise unremarkable — roughly uniformly spread over the acceptance (Fig. 2b), i.e. no special beam/telescope location.

**The "no clean track" population (a readout confound).** 60 % of all sparks fall in events where M3 finds *no* clean track, versus a 33 % baseline — sparks are 2× over-represented there, and those sparks are the *same size* as the tracked ones (median 91 vs 95 strips). The most likely reading is that a large discharge injects common-mode noise into the *shared* DAQ and spoils M3 track reconstruction for that event (or that both are triggered by a messy multi-particle event). Either way it is a selection/cross-talk artefact, not evidence of a large muon-independent spark source, so the clean 4.9 % vs 1.2 % comparison above is restricted to good-track events. *This DAQ cross-talk is itself worth a follow-up.*

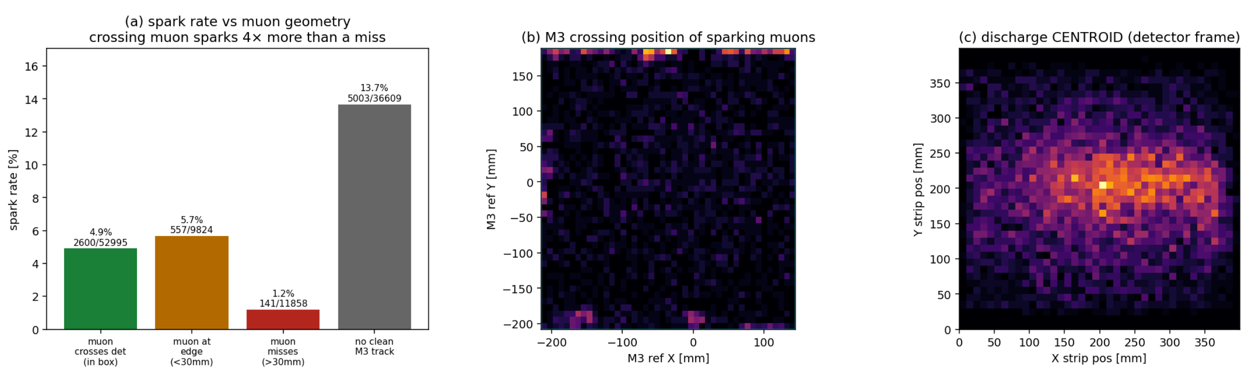


Figure 2: **Muon dependence.** (a) spark rate split by where the muon projects — crossing muons spark 4× more than missing ones; (b) M3 crossing positions of sparking muons (roughly uniform); (c) discharge centroid in the detector frame (central).

# 5 Q2b — Are the “no-track” sparks really particle showers? No

A natural competing explanation for the no-track population is a **shower**: a burst of several particles that crosses both M3 (leaving a mess that fails to fit a single track) and our detector (firing many strips, read as a spark). This is directly testable, because a shower of several charged particles leaves M3 with *multiple straight-track candidates* (ncand $\geq 2$), whereas a corrupted or particle-free event leaves ncand $= 0$ (M3 blank) or a single degraded candidate (ncand $= 1$). We count raw M3 track candidates per event (before quality cuts) and split by spark class (Fig. 3): The

| event class | ncand $= 0$ (blank) | ncand $= 1$ (single) | ncand $\geq 2$ (**shower**) |
|---|---|---|---|
| non-spark | 17.5 % | 82.3 % | 0.15 % |
| all sparks | 34.1 % | 64.9 % | 1.02 % |
| **no-track sparks** | 56.5 % | 42.2 % | **1.26 %** |

shower signature is essentially absent: only **1.3 %** of no-track sparks have $\geq 2$ M3 tracks (det7: 0.5 %), and the M3 tracker never returns more than three candidates — multi-track events are 0.2 % of the entire run, far too rare to account for the thousands of no-track sparks. Instead the no-track sparks are M3-blank (57 %) or a single track that failed the quality cut (42 %). Second, the **discharge morphology is identical** for muon-tracked and no-track sparks (median 88 vs 91 strips, span 304 vs 372 mm, time spread 251 vs 257 ns, edge and saturation fractions equal; Fig. 3c) — the no-track sparks are the *same physical discharge* as the muon-induced ones, not a distinct multi-cluster shower signature. **The shower hypothesis is ruled out.**

What the split *does* show is that a spark nearly **doubles** the chance that M3 returns no track at all (17.5 → 34 % blank): the discharge suppresses M3 reconstruction (the DAQ common-mode cross-talk), on top of a genuinely muon-free spontaneous component. Cleanly separating those two would need the M3 *raw hits* (not stored in the ray files) or a spark↔M3-noise timing coincidence — a further step — but showers are not the answer.

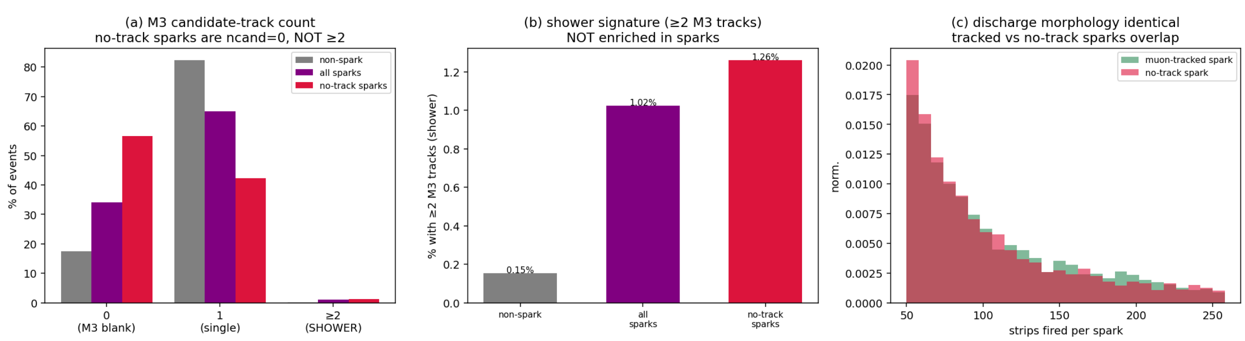


Figure 3: **Shower test.** (a) M3 candidate-track count by spark class — no-track sparks are ncand $= 0/1$, not $\geq 2$; (b) the $\geq 2$-track shower fraction is not meaningfully enriched in sparks; (c) muon-tracked and no-track sparks have identical discharge size (and span/duration/edge — see text).

# 6 Q3 — Are there particular places that spark? Yes: the edges

Strip occupancy in sparks is strongly non-uniform (Fig. 4a,b). Both planes show large peaks at the detector **edges** (0 and ∼399 mm; strongest at high-$X$ and low-$Y$), where field lines concentrate at the active-area boundary — edge strips participate in a large fraction of all sparks. On top of the edge structure the $Y$ plane (FEU8) shows a handful of narrow spikes at fixed positions — individual **hot strips** that fire far above average. Yet the discharge *centroid* sits centrally (Fig. 2c): a typical spark ignites over a ∼200 mm span (median $X$ span 204 mm, $Y$ span 235 mm of the 399 mm plane) — roughly half the detector, not a point and not the whole plane — with the edges over-weighted. The $Y$/FEU8 plane discharges harder than $X$ (median 53 vs 40 strips; Fig. 4c), consistent with the FEU8 saturation band seen in the reco_far study. Only 10 % of spark hits actually saturate the ADC, so these are genuine spread discharges, not just electronic rail-outs.

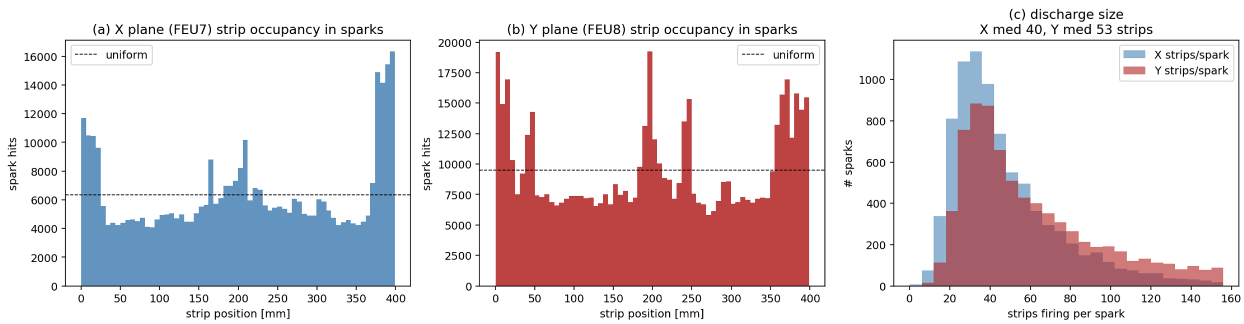


Figure 4: **Where.** (a,b) strip occupancy in sparks for the X and Y planes vs a uniform baseline — edge peaks (both planes) and fixed hot strips (Y); (c) strips firing per spark — Y discharges are larger than X.

## 7 Spark hitmaps — and a second, detector-side shower check

Fig. 5 maps where sparks live in the detector plane. The discharge *centre* (mean of a spark's fired strips) is a central blob (a), but the *footprint* hitmap (b) — the fraction of sparks with coincident X & Y activity in each cell — is a striking fixed **grid**: bright bands along all four edges plus a few fixed hot strips, the same cells lighting up spark after spark, over a dark (∼5 %) interior. This fixed footprint is itself a shower veto: a shower would land where the muon flux does (uniformly, as in Fig. 2b), *not* repeatedly on the same edges and hot channels. The edges/hot-strips are a detector property.

The fired-strip micro-structure (Fig. 6) says the same thing from the mx17 side, addressing the concern that a *tight* shower (which M3 could not resolve) might mimic a spark. A spark is a few broad blobs (median 4 contiguous clusters per plane, patchy fill ∼0.19) spread over ∼half the plane — but the crucial number is the **widest contiguous cluster: median 19 mm** (Fig. 6c), with no population at the ∼5–8 mm width of a single-particle track. Real particle tracks, however tightly bunched, would show up as ∼mm-wide clusters; the spark blobs are 2–4× wider and sit at fixed positions. So even a collimated shower is excluded: the detector sees discharge blobs, not tracks.

## 8 Q4 — All strips at once, or does the discharge "walk"? Neither

The discharge is *not* a single simultaneous flash: only 1.2 % of sparks have all their hits within 100 ns; the per-spark hit-time spread is $\sigma \approx 253$ ns median (tail beyond 500 ns), somewhat broader than a normal drift cluster (193 ns, Fig. 7a). But it does *not* propagate spatially either: the per-spark position–time correlation is $|r| \approx 0.12$ (only 3 % exceed 0.5), and the spark-centred Δposition–Δtime map (Fig. 7b) is a clean **cross with no tilt** — a horizontal arm (strips across the whole span firing near-simultaneously) plus a vertical arm (the core keeps firing over ±hundreds of ns), and no diagonal that a moving front would produce. Example events (Fig. 7c) show strips

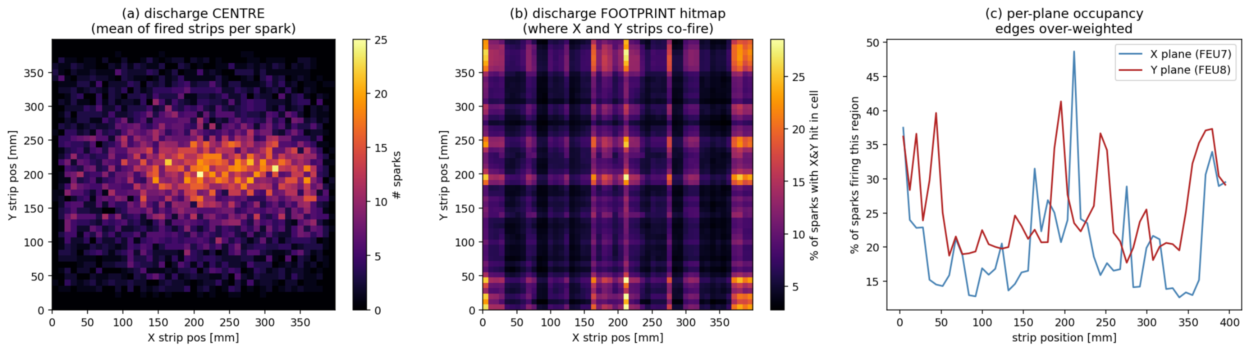


Figure 5: **Spark hitmaps** (detector strip frame). (a) discharge centre (mean of fired strips per spark); (b) footprint — % of sparks with X & Y activity per cell, a fixed edge+hot-strip grid; (c) per-plane occupancy, edges over-weighted.

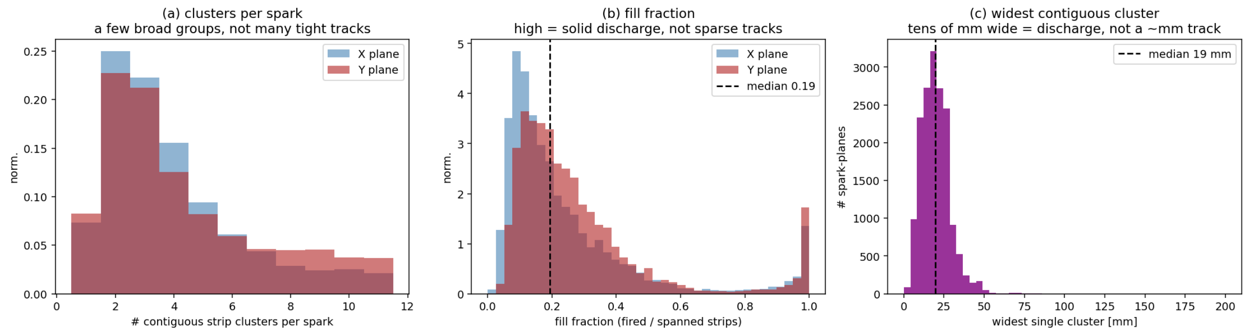


Figure 6: **Detector-side shower check.** (a) contiguous clusters per spark; (b) fill fraction; (c) widest contiguous cluster — median 19 mm, far wider than a ∼mm particle track, so the fired strips are discharge blobs, not tight tracks.

lighting up across the full 0–400 mm range at scattered times. **Picture:** a spatially-extended discharge that fires a broad region in a near-synchronous burst and then rings/afterpulses for up to ∼1 µs, with timing independent of position — not a wave sweeping across the strips.

# 9 Summary and follow-ups

- **Time:** Poisson-random, stable 0.33 Hz — not HV bursting.
- **Cause:** predominantly **muon-induced** — a crossing muon sparks 4× more than a missing one (4.9 % vs 1.2 %) — on a ∼1.2 % spontaneous floor. Lowering the resist HV reduces the *rate* (fewer avalanches reach breakdown) but cannot remove the muon-seeded component entirely.
- **Place: edge-dominated** occupancy plus fixed hot strips (Y/FEU8); discharges span ∼half the plane, centred centrally; Y sparks harder than X.
- **Structure:** a spatially-extended, temporally-spread (∼250 ns, tail to ∼1 µs) discharge with afterpulsing — no propagating "walk".
- **Not showers:** the "no-track" sparks are not multi-particle showers — only 1.3 % carry $\geq 2$ M3 tracks and their discharge morphology is identical to muon-tracked sparks (Q2b). The lost M3 track is DAQ cross-talk + spontaneous muon-free sparks, not a physics shower.

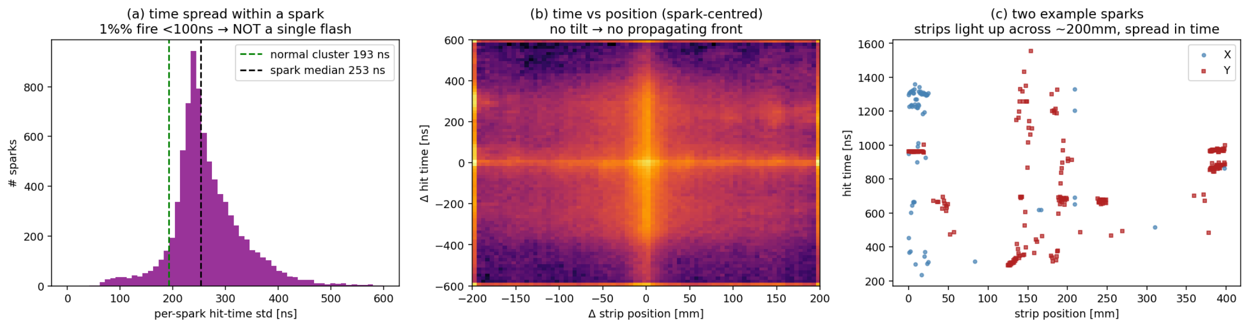


Figure 7: **Time structure.** (a) per-spark hit-time spread vs a normal cluster; (b) spark-centred time vs position (no tilt ⇒ no propagating front); (c) two example sparks — strips across ∼400 mm, spread in time, no position ramp.

- **Follow-ups:** (1) inspect the edge / guard-ring field and the Y/FEU8 hot strips (mask or lower gain); (2) confirm and mitigate the **DAQ common-mode cross-talk** by which a det3 spark spoils M3 tracking (a spark doubles M3's no-track rate, 17.5 → 34 %) — next step is M3 raw hits or a spark–M3-noise timing coincidence; (3) an offline spark tag (multiplicity + the ∼1 µs duration signature) would clean the reco_far tail (companion note `det3_recofar_analysis`).

*Reproduce:* `extract_sparks.py` then `make_plots.py`; numbers in `spark_meta.json`. Spark cut (>50 strips) and alignment match `09_efficiency_breakdown.py`.